

Extraction of Microservices from Monolithic Software Architectures

Source: G. Mazlami *et al.* "Extraction of Microservices from Monolithic Software Architectures," 2017 *IEEE International Conference on Web Services (ICWS)*, 2017. [Online]. Available: <http://ieeexplore.ieee.org/document/8029803/>

Paper Information: Key Paper

Rating: ★★★★★

Own Keywords: Approaches, Clustering, Graph-based

Reading Progress: Second Pass

Introduction (What is the Paper about?)

- addressing the decomposition problem of monolithical applications into microservices **algorithmically and graph-based**
- apply approaches from the software decomposition research formally and (semi-)automated to microservices

Methods

- constructed a web-based prototype
- tested three formal coupling strategies used to construct a graph from code
 - perform a clustering algorithm on that
- performance evaluation on the prototype with custom microservice-specific metrics on 21 open source projects in Java, Ruby and Python

Scientific Benefit

- Presenting a formal microservice extraction model as web-based prototype
 - semi-automated without heavy user input
- allow algorithmic recommendations of microservice candidates in a refactoring and migration scenario
- size of produced microservice candidates conforms with reported microservice size in empirical surveys
- produced microservice candidates lower the average development team size down to half of the original size or lower
- domain-specific redundancy among different microservices is kept low

Challenges / Future Work

Limitations

- extraction model uses classes as atomic unit rather than methods or procedures
 - left for future work
- extraction model treats database entities as normal classes
 - still unsolved to share or assign pre-existing databases to different models

Paper Outline and Read Progress

1. ☒ Introduction
2. ☒ Related Work
3. ☐ Formal Definition of the Extraction Models, Coupling Strategies and Clustering Algorithm
4. ☐ Performance and Quality Evaluation
5. ☒ Limitations and Future Work
6. ☐ Conclusion

Own Comments

- (+) Potential key paper
- (+) First Paper applying software decomposition methods to microservices
- (-) potentially outdated
- (-) custom metrics
- (-) missing reference of Komondoor et al.